CSE 421 ID: 23101510

Application Layer Protocols (HTTP, SMTP/POP)

Examination Lab — Answer Sheet

Objectives: Capture traffic and observe the PDUs for HTTP, SMTP, POP3.

Task 1: Observe HTTP traffic exchange between a client and server

Packet 1

Q: What kind of HTTP packet is packet no. 1?

Ans: Packet 1 is an HTTP Request packet. Specifically, it is a GET request that PC1 (the client) sends to the local web server, asking the server to return the web page located at www.bracu.ac.bd.

Q: Click onto "Inbound PDU details" tab. Scroll down at the end, what do you see?

Ans: The HTTP header part is displayed at the bottom of the Inbound PDU Details under Layer 7 (Application layer). Before it is sent to the transport layer (TCP) for transmission, it confirms that the PC has generated an HTTP GET request by displaying the request line, which includes the HTTP method (GET), the requested URL/host, and the HTTP version (HTTP/1.1).

Packet 2

Q: Click onto "Inbound PDU details" tab. Scroll down at the end, what do you see? What kind of HTTP packet is this?

Ans: The HTTP Response header, which includes the HTML content of the requested page after a status line like "HTTP/1.1 200 OK," can be seen by scrolling to the bottom of Layer 7 for this packet. This verifies that packet 2 is an HTTP response, which is the web server's response that sends PC1 the requested webpage.

Task 2: Observe email traffic exchange between a client and email server using SMTP and POP3

Step 2 – SMTP capture

Packet 4

Q: What is the purpose of this DNS packet?

Ans: This DNS packet is sent by PC1 to resolve the domain name of the Bracu Email Server (e.g. mail.bracu.ac.bd) into its corresponding IP address, since PC1 needs the server's IP address before it can establish an SMTP connection to send the email.

Packets 5 & 6

Q: Explain why an SMTP packet was sent to the email server and the server replied with an SMTP packet.

Ans: The protocol used to send or upload outgoing mail from a client to a mail server is called SMTP (Simple Mail Transfer Protocol). In order for the email to be forwarded or saved for delivery, PC1 transmits an SMTP packet with the sender, recipient, subject, and body to the Bracu Email Server. In order to acknowledge each step and verify that the email was successfully received, the email server processes the SMTP commands (such as HELO/EHLO, MAIL FROM, RCPT TO, and DATA) and

returns an SMTP reply packet with status/response codes (such as 220 Ready, 250 OK, 354 Start mail input, and 250 Message accepted).

Step 3 – POP3 capture

Packet 6

Q: What is the purpose of this DNS packet?

Ans: DNS packet resolves the domain name of the Bracu Email Server into its IP address this time so that PC1knows where to send the POP3 request to retrieve messages from the mail server.

Packets 7 & 8

Q: Explain why a POP packet was sent to the email server and the server replied with a POP packet.

Ans: POP3 is the protocol used by a client to download email that is waiting in its mailbox on the server. In order to log in and retrieve any stored messages, the client submits a POP3 request to the Bracu Email Server. The exchange looks as a request POP3 packet followed by a reply POP3 packet because the server responds with POP3 response codes and, upon successful authentication, transmits the stored email content back to the client.